```c
#include <stdio.h>
#include <string.h>

#define MAX_HISTORY 5
#define MAX_LENGTH 100

int main(void) {
    char history[MAX_HISTORY][MAX_LENGTH];
    char input[MAX_LENGTH];
    int count = 0;

    while (1) {
        printf("\nEnter command (history / exit to stop): ");

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        if (strcmp(input, "history") == 0) {
            printf("\nCommand History:\n");

            int start = count > MAX_HISTORY ? count - MAX_HISTORY : 0;

            for (int i = start; i < count; i++)
                printf("%d. %s\n", i + 1,
                       history[i % MAX_HISTORY]);

            if (count == 0)
                printf("History is empty.\n");

            continue;
        }
```

```
krishnasree08@krishnasree08:~$ mkdir -p ~/ossp_session11_12
cd ~/ossp_session11_12
krishnasree08@krishnasree08:~/ossp_session11_12$ nano command_history.c
krishnasree08@krishnasree08:~/ossp_session11_12$ gcc -Wall -Wextra command_h
istory.c -o command_history
./command_history

Enter command (history / exit to stop): exit
Program ended.
```

```
krishnasree08@krishnasree08:~/ossp_session11_12$ nano pipeline_structure.c
krishnasree08@krishnasree08:~/ossp_session11_12$ gcc -Wall -Wextra pipeline_
structure.c -o pipeline_structure
./pipeline_structure
Pipeline Structure Builder
Enter commands one by one.
Type done to finish.
Command 1: cat
Command 2: grep main
Command 3: wc -1
Command 4: done

Pipeline Layout:
cat | grep main | wc -1
Total commands: 3
Pipeline structure created successfully.
```

```
krishnasree08@krishnasree08:~/ossp_session11_12$ nano pipe_execution.c
krishnasree08@krishnasree08:~/ossp_session11_12$ gcc -Wall -Wextra pipe_exec
ution.c -o pipe_execution
./pipe_execution
6
Pipe execution completed.
krishnasree08@krishnasree08:~/ossp_session11_12$ nano multiple_pipes.c
krishnasree08@krishnasree08:~/ossp_session11_12$ gcc -Wall -Wextra multiple_
pipes.c -o multiple_pipes
./multiple_pipes
4
Multiple pipe execution completed.
```

Overall Result:

Thus, the OSSP programs were implemented to demonstrate command history management, pipeline structure creation, inter-process communication using pipes, execution of multiple connected commands, process synchronization, stream redirection, and proper resource management.